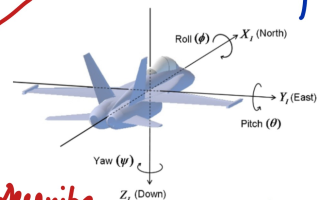
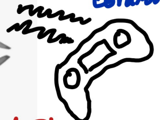
Drone Control Loop

IMU → Accelerometer, Gyroscope → Sensor Fusion Algorithm → Euler Angles → Yaw, Pitch, Roll → PID Controller → Flight Controller



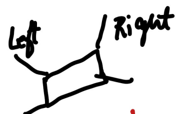
Euler Angles → describe 3D orientation of object with respect to fixed reference frame

* Throttle : The current, real-time number the Remote Controller (RC) sends the Flight Controller (FC), comes from Joystick.
Let's say range from 0 (motor off) to 1000 (full power)



* Random Internet Pic

Example:



Drone Flying → Wind gust → Unstable! → PID

To stabilize : Push left up, Drop right down

↳ Stable!

Leftmotors spin faster, Right slower

So motor_left = throttle + roll correction
motor_right = throttle − roll correction

} Assuming all motors spin in the same direction

Motor PWM

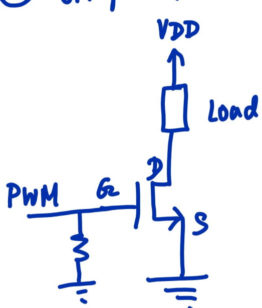
① Using Electronic Speed Controllers (ESCs) → expect a pulse every 20ms.

* Pulse of 1ms = 0% Throttle
* Pulse of 2ms = 100% Throttle

So, take throttle (0 − 1000) ⟶ map to pulse width (1 ms − 2 ms) → Set PWM pulse

External ESC → may add weight!

② Simple low side switching (+ Flyback Diode connected across motor load (inductive) to prevent damage from voltage spikes)



So, can map motor_left & motor_right directly to PWM percentage

* Note: Choose high PWM frequency to reduce vibration (≈ 20 kHz)